

February 11th 2025

Graphene synthesis project update for previous week

Objective 1: Production of 300 g of biochar

MB3046

Objective: Test-reaction with milled biochar and unmilled KOH flakes

Experiment details

Biochar (MB3032) milled (Blendtec, 90 sec) => 16.1 g powder

Reaction in rotating oven, 800 °C, 1 h, 3 °C/min, 15.6 g powder and 24.2 g KOH flakes loaded, 7.05 g (90 ml) output

Result: Some species 1, some species 2, mostly small, solid particles, normal yield

Conclusion: Output different from "normal" process with milled KOH

Recommended action: Evaluation of product properties

MRa385

Objective: Test milling in 15 sec intervals (90 sec total) to prevent/limit heating of material during milling

Experiment details

18.9 g biochar (MRa367) milled (Blendtec, 6x15 sec) with KOH (1:1.5), unloaded in glove box -> brown powder, no change in appearance during bottling, occurrence of dark material on blender walls not improved (see photo)

Rotating oven, 800 °C, 1 h, 3 °C/min, 44.5 g powder used, 8.6 g (125 ml) output

Result: Typical yield, product volume increased, mostly species 1/2 mix, some smooth surfaces

Conclusion: Benefit of short milling intervals unclear, increased ratio of species 2 consequence of using air-dried biochar?

Recommended action: Repeat experiment with direct comparison to non-interval milling using same biochar lot



Overall status

- 6 rotating oven batches completed, total 45 g product
- max batch size not reached
- oven setup is being refined (latest addition: redesign of gas scrubber)

- product quality consistency moderate, blender performance and biochar lot variability suspected cause for quality and consistency deficits
- product density variable (8-15 ml/g)
- current frequency is two batches (approx. 15 g product) per week, test of increase in progress
- step 1 production on lab scale in progress, three batches (approx. 60 g) per week, pilot plant production pending, to be started next week, lab scale production to be stopped
- So far, products only characterized by SEM

Objective 2: Testing/Optimization on small scale

MRa380

Objective: Small-scale test with increased water content (approx. 11.5%), to test whether this leads to higher selectivity for species 1

Experiment details

3.0 g biochar (MB2996) treated with 156 mg water ($\Rightarrow$ KFT 7.7%), then with 157 mg water ($\Rightarrow$ KFT 11.5%), then 2.0 g milled with KOH (1:1.5, 3 min) $\rightarrow$ brown powder, no change in appearance during bottling

800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven A), 480 mg (5.0 ml) output

Result: Product is mostly species 1/2 mix, some smooth surfaces

Conclusion: At this moisture level no improvement was observed, species 1 ratio in product actually decreased

Recommended action: Conclude water addition experiments

MRa386

Objective: Second test of 1:1 ratio (biochar/KOH) on small scale, slightly larger scale to take advantage of better volume efficiency

Experiment details

3.5 g biochar (MRa368) milled with KOH (1:1, 3 min) $\rightarrow$ brown powder, no change in appearance during bottling

800 °C, 1 h, 3 °C/min, 2 x 2.1 g pellets pressed, standard inertion (oven A), some spillage during reaction, 653 mg (3.7 ml) output

Result: Lower-than-average yield (32%, similar to previous 1:1 experiment MB3041) volume increase during reaction higher than expected (leading to some spillage), pure/uniform species 1, pore size normal (unlike MB3041)

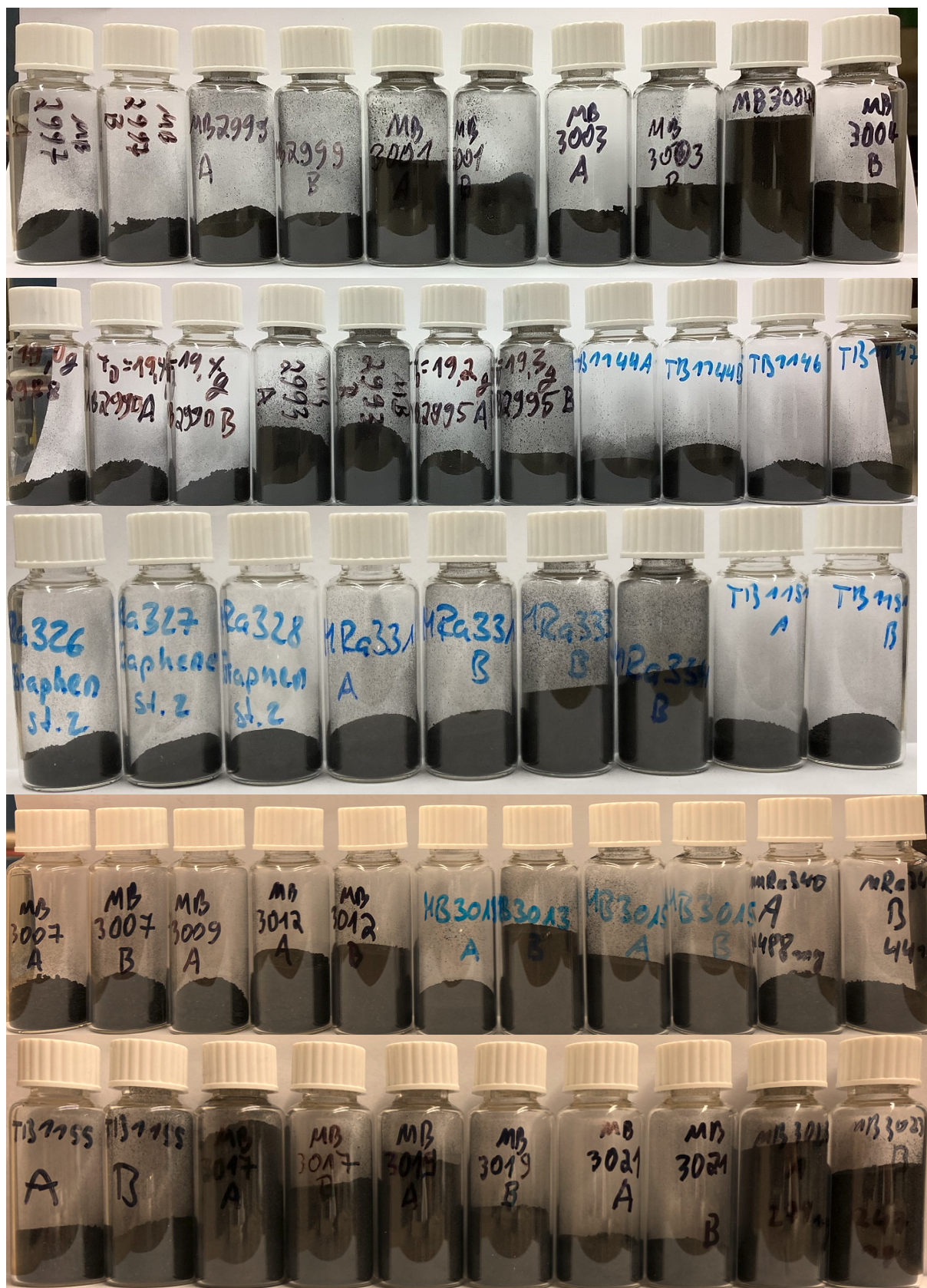
Conclusion: 1:1 ratio generally feasible, low yield confirmed, changes in volume efficiency and pore size not confirmed

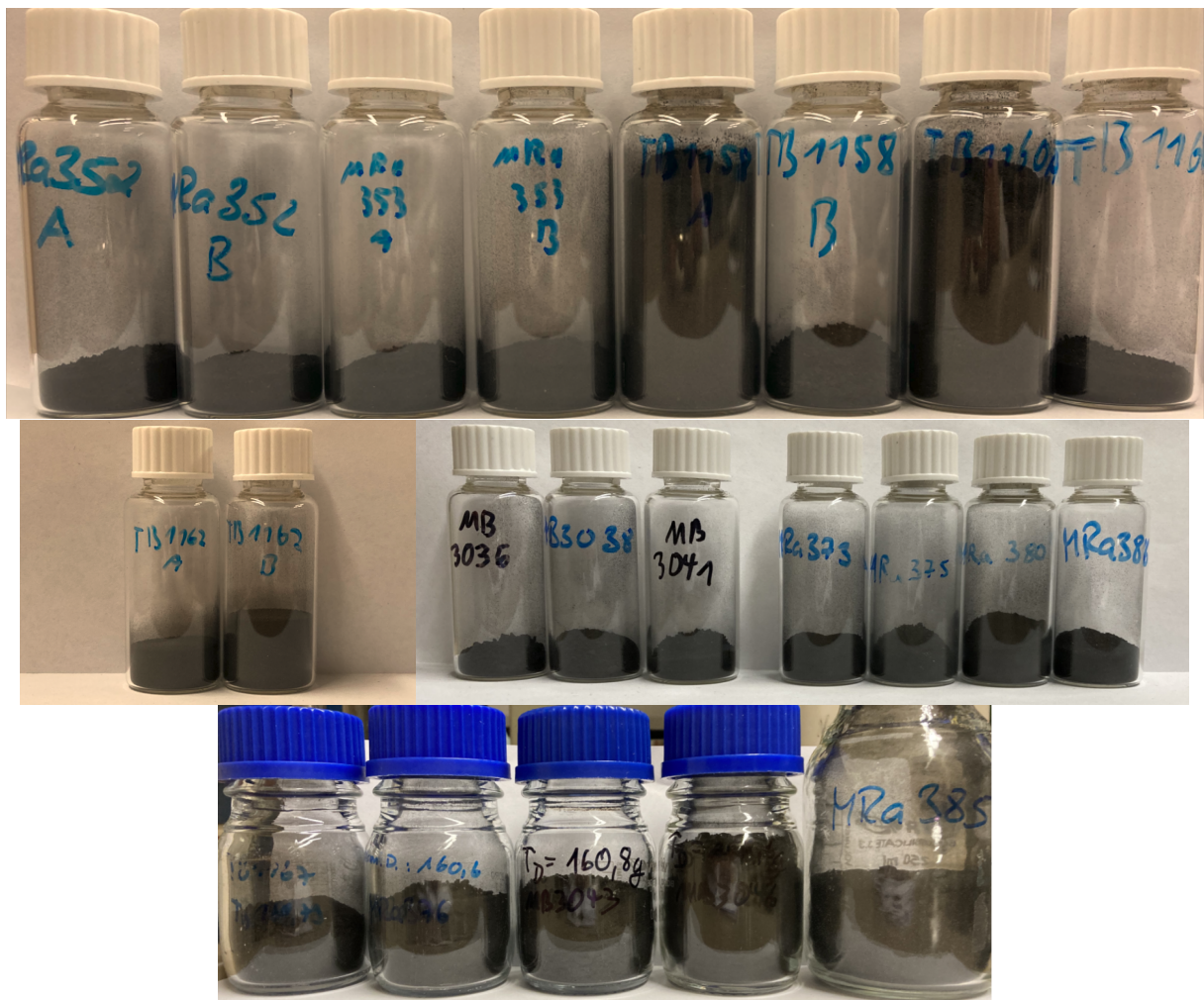
Recommended action: Follow up with larger batch (blender/rotating oven)

Overall status

- Increased water content: Not productive, no clear correlation between moisture content and product appearance, certainly no preferred formation of species 1
- 1:1 ratio (biochar/KOH): Lower yield, higher homogeneity, likely better volume efficiency, to be followed up on with further experiments

Product pictures





Analytical data

- SEM (and partially Raman) results available on sharepoint for

- TB1133, MB2946A, MB2947D, TB1135-1, TB1135-2, MB2952B, MB2952C, MB2952C2, MB2955A, MB2955A2, TB1136, MB2962A/B, MB2963A/B, TB1137-1/2/3 (step 2 products), MB2965B, MB2965B2, MB2966A/B, MB2967A/B, MB2970A/B, MB2971A/B, MB2972A/B, MB2973A/B, MB2974A/B, MB2975A/B, MB2976A/B, MB2979A/B, MB2980A/B/C/D, KJo-0165A/B, KJo-0166A/B, KJo-0167A/B, KJo-0168A/B, KJo-0169, KJo-0170, KJo-0171A/B, KJo-0172A/B, KJo-0175, KJo-0176, KJo-0177, KJo-0178, KJo-0171A GR, MB2981A, TB1138, MB2984, MRa320, TB1139, TB1140A, TB1140B, MB2988, MRa323, TB1141, MRa326, MRa327, MRa328, MB2990A, MB2990B, MRa329, MRa331A, MRa331B, MRa333A, MRa333B, MRa334A, MRa334B, MRa2993A, MRa2993B, MB2995A, MB2995B, MB2997A, MB2997B, TB1144A, MB1144B, MB2999A, MB2999B, TB1146, TB1147, MB3001A, MB3001B, MB3003A, MB3003B, MB3004A, MB3004B, TB1151A, TB1151B, MB3007A, MB3007B, MB3009, MB3013A, MB3013B, MB3015A, MB3015B, MRa340A, MRa340B, MB3017A, MB3017B, MB3019A, MB3019B, MB3021A, MB3021B, MB3023A, MB3023B, MRa352A, MRa352B, MRa353A, MRa353B, TB1158A, TB1158B, TB1142, TB1160A, TB1160B, TB1162A, TB1162B, MB3026, MB3027, MB3028, MB3030, TB1165, TB1166, MB3036, MB3038, MRa373, TB1167, MRa375, MB3041, MRa376, MB3043, MRa380, MRa385, MRa386, MB3046 (step 2 products)
- 900561-500MG, 2 lots (Sigma-Aldrich reference)
- 924458-500MG (Sigma-Aldrich reference "Single-layer graphene sheets for battery, Bio-sourced")
- Graphene oxide

- TEM results available for

- MB2955A2 (step 2 product)
- MB2976A (step 2 product)
- TB1137-1 (step 2 product)

- BET results

Sample	Multipoint BET Area [m ² /g]	Langmuir Surface Area [m ² /g]	Species
MB2976A	1.88 x 10 ³	2.55 x 10 ³	mostly 1
MRa329	1.76 x 10 ³	2.39 x 10 ³	2
MRa333A	2.09 x 10 ³	5.74 x 10 ³	2
MRa334A	1.86 x 10 ³	2.50 x 10 ³	mostly 2
MB3001A	1.613 x 10 ³	7.763 x 10 ³	mostly 2
MB3004A	1.491 x 10 ³	2.032 x 10 ³	mostly 2
MB3001A/3004A	1.471 x 10 ³	1.983 x 10 ³	mostly 2
924458-500MG	1.112 x 10 ³	1.421 x 10 ³	n.a.
900561-500MG	1.942 x 10 ¹	2.442 x 10 ¹	n.a.
TB1142	1.257 x 10 ³	1.684 x 10 ³	1
TB1160B	1.240 x 10 ³	1.677 x 10 ³	1
TB1165	1.520 x 10 ³	2.020 x 10 ³	1

Experiment overview – Step 1

Experiment	Reactor	Raw material		Acid				T	t	pressure		Wash		Output		
MB2928	AV1	6.0 g	BAFA neu Hemp Fibre VF	100.0 g	0.19%	0.02 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	-> 10.4 bar	742 g	Water	2.6 g		
Comment	Fibres not completely covered by liquid															
MB2933	AV1	6.1 g	BAFA neu Hemp Fibre VF	174.0 g	0.19%	0.02 M	Sulfuric acid	180 °C	24 h	8.7 bar	-> 9.5 bar	647 g	Water	1.2 g		
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2935	AV1	6.0 g	BAFA neu Hemp Fibre VF	172.9 g	0.19%	0.02 M	Sulfuric acid	180 °C	23 h	8.8 bar	-> 9.5 bar	500 g	Water	1.3 g		
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2936	AV1	6.0 g	BAFA neu Hemp Fibre VF	172.4 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h	8.8 bar	-> 10.6 bar	589 g	Water	0.8 g	Combined and homogenized as lot no MRa231	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa225	AV1	6.0 g	BAFA neu Hemp Fibre VF	153.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	?	-> 10.5 bar	500 g	Water	0.9 g		
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa228	AV1	6.3 g	BAFA neu Hemp Fibre VF	154.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	-> 11.0 bar	500 g	Water	0.9 g		
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa231	AV1	6.1 g	BAFA neu Hemp Fibre VF	168.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.0 bar	-> 10.5 bar	500 g	Water	0.8 g		
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2937	AV1	9.4 g	Canadian Rockies Hemp	179.3 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar	-> 10.9 bar	569 g	Water	1.6 g	Combined and homogenized as lot no MB2946X	
Comment	very low amount of foreign material observed and removed during filtration															
MB2938	AV1	8.9 g	Canadian Rockies Hemp	175.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	9.2 bar	-> 13.2 bar	577 g	Water	1.4 g		
Comment	very low amount of foreign material observed and removed during filtration															
MB2940	AV1	9.2 g	Canadian Rockies Hemp	177.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	-> 10.6 bar	555 g	Water	1.5 g		
Comment	very low amount of foreign material observed and removed during filtration															
MB2942	AV1	8.9 g	Canadian Rockies Hemp	177.6 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h	8.3 bar	-> 13.5 bar	578 g	Water	1.6 g		
Comment	low amount of foreign material observed and removed during filtration															
MB2943	AV1	9.8 g	Canadian Rockies Hemp	178.2 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.3 bar	-> 12.7 bar	611 g	Water	1.7 g		
Comment	low amount of foreign material observed and removed during filtration															
MB2944	AV1	9.0 g	Canadian Rockies Hemp	178.4 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.4 bar	-> 9.7 bar	593 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration															
MB2945	AV1	9.1 g	Canadian Rockies Hemp	177.7 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.7 bar	-> 9.8 bar	6 g	Water	1.6 g		
Comment	low amount of foreign material observed and removed during filtration															
MB2948	AV1	9.4 g	Canadian Rockies Hemp	175.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	25 h	9.0 bar	-> 12.6 bar	475 g	Water	1.7 g		
Comment	low amount of foreign material observed and removed during filtration															

Experiment	Reactor	Raw material		Acid			T	t	pressure		Wash		Output	Drying	KFT
MB2949	AV1	9.2 g	Canadian Rockies Hemp	174.3 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.4 bar -> 10.2 bar	538 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2950	AV1	9.1 g	Canadian Rockies Hemp	170.5 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h	-> 11.1 bar	534 g	Water	1.6 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2951	AV1	9.3 g	Canadian Rockies Hemp	178.5 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	9.1 bar -> 11.2 bar	575 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration														
MRa255	AV1	9.2 g	Canadian Rockies Hemp	170.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.1 bar -> 12.8 bar	488 g	Water	1.7 g		
Comment	some foreign material observed and removed during filtration														
MB2953	AV1	9.7 g	Canadian Rockies Hemp	177.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	9.1 bar -> 10.4 bar	558 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2956	AV1	9.8 g	Canadian Rockies Hemp	172.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar -> 12.5 bar	548 g	Water	1.9 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2957	AV1	9.1 g	Canadian Rockies Hemp	176.8 g	1.50%	0.15 M	Sulfuric acid	180 °C	23.5 h	8.8 bar -> 8.5 bar	589 g	Water	1.5 g		
Comment															
MB2958	AV1	9.3 g	Canadian Rockies Hemp	177.0 g	2.00%	0.20 M	Sulfuric acid	180 °C	24 h	8.9 bar -> 8.7 bar	562 g	Water	1.8 g		
Comment															
MB2959	AV5	76.0 g	Canadian Rockies Hemp	953.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	25 h		3957 g	Water	14.9 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2960	AV1	3.8 g	Canadian Rockies Hemp (cut)	151.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h	8.8 bar -> 10.5 bar	300 g	Water	0.6 g		
Comment	Reaction performed with stirring														
MB2961	AV1	9.9 g	Canadian Rockies Hemp	172.0 g	5.00%	0.51 M	Sulfuric acid	180 °C	24 h	10.1 bar -> 16.1 bar	563 g	Water	1.7 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2964	AV1	9.0 g	Canadian Rockies Hemp	211.0 g + 140.8 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar -> 8.8 bar	310 g + 603 g	Water	1.3 g		
Comment	Hemp boiled with 1% H2SO4 at 95 °C, filtered and washed, then heated in Autoclave with fresh acid														
MB2978	AV5	76.3 g	Canadian Rockies Hemp	822.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	22 h		4112 g	Water	15.4 g		
Comment															
KJo-0173	AV5	77.1 g	Canadian Rockies Hemp	942.1 g	1.00%	0.10 M	Sulfuric acid	185 °C	24 h		4283 g	Water	16.5 g	40 °C	4.9%
Comment														100 °C	2.3%
KJo-0174	AV5	77.1 g	Canadian Rockies Hemp	945.0 g	1.00%	0.10 M	Sulfuric acid	190 °C	24 h		4384 g	Water	15.8 g	40 °C	3.7%
Comment															
MB2982	AV5	77.1 g	Canadian Rockies Hemp	999.5 g	1.00%	0.10 M	Sulfuric acid	190 °C	24 h		4258 g	Water	14.9 g	40 °C	5.3%
Comment															
MB2983	AV5	80.5 g	Canadian Rockies Hemp	999.8 g	1.00%	0.10 M	Sulfuric acid	190 °C	24 h		4232 g	Water	16.6 g	40 °C	5.5%
Comment														100 °C	1.7%

Experiment	Reactor	Raw material	Acid	T	t	pressure	Wash	Output	Drying	KFT
MB2985	AV5	79.5 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	200 °C	24 h		4241 g Water	15.4 g	40 °C	3.7%
Comment									100 °C	1.9%
MB2986	AV5	79.9 g Canadian Rockies Hemp	999.9 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4128 g Water	16.0 g	60 °C	2.7%
Comment									100 °C	1.7%
MB2989	AV5	80.2 g Canadian Rockies Hemp	999.3 g 1.00% 0.10 M Sulfuric acid	180 °C	23 h		4314 g Water	18.2 g	40 °C	5.9%
Comment										
MB2991	AV5	81.3 g Canadian Rockies Hemp	999.3 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4223 g Water	19.1 g	40 °C	5.7%
Comment										
MB2992	AV5	81.5 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	160 °C	23.5 h		4262 g Water	19.5 g	40 °C	6.2%
Comment										
MB2994	AV5	80.8 g Canadian Rockies Hemp	999.8 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4159 g Water	20.2 g	40 °C	5.2%
Comment										
MB2996	AV5	82.7 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4210 g Water	20.0 g	40 °C	4.6%
Comment										
MB2998	AV5	84.0 g Canadian Rockies Hemp	999.6 g 1.00% 0.10 M Sulfuric acid	180 °C	22 h		4263 g Water	19.8 g	40 °C	5.0%
Comment	Included nickel and steel material samples for corrosion/cleaning tests									
MB3000	AV5	82.5 g Canadian Rockies Hemp	1000.0 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4431 g Water	19.8 g	40 °C	5.0%
Comment										
MB3002	AV5	80.0 g Canadian Rockies Hemp	999.9 g 1.00% 0.10 M Sulfuric acid	190 °C	25 h		4245 g Water	18.5 g	40 °C	2.0%
Comment										
MB3005	AV5	81.4 g Canadian Rockies Hemp	999.4 g 1.00% 0.10 M Sulfuric acid	200 °C	24 h		4280 g Water	19.0 g	40 °C	2.0%
Comment										
MB3006	AV5	80.8 g Canadian Rockies Hemp	1000.1 g 1.00% 0.10 M Sulfuric acid	190 °C	23 h		4182 g Water	17.4 g	40 °C	2.1%
Comment										
MB3008	AV5	82.2 g Canadian Rockies Hemp	1000.2 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4238 g Water	18.0 g	40 °C	2.6%
Comment										
MB3010	AV5	81.3 g Canadian Rockies Hemp	1000.0 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4235 g Water	18.9 g	40 °C	2.0%
Comment										
MB3011	AV5	82.4 g Canadian Rockies Hemp	999.4 g 1.00% 0.10 M Sulfuric acid	160 °C	24 h		4244 g Water	23.5 g	40 °C	2.2%
Comment										
MB3014	AV5	82.8 g Canadian Rockies Hemp	999.8 g 1.00% 0.10 M Sulfuric acid	160 °C	23 h		1460 g Water	23.6 g	40 °C	1.3%
Comment										
MRa341	AV5	82.3 g Canadian Rockies Hemp	1000.4 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		1232 g Water	19.2 g	40 °C	2.1%
Comment										

Experiment	Reactor	Raw material	Acid	T	t	pressure	Wash	Output	Drying	KFT
MB3016	AV5	81.7 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	185 °C	21 h		4232 g Water	19.7 g	40 °C	2.1%
Comment										
MB3018	AV5	81.2 g Canadian Rockies Hemp	1000.1 g 1.00% 0.10 M Sulfuric acid	195 °C	22 h		4266 g Water	19.0 g	40 °C	2.1%
Comment										
MB3020	AV5	81.1 g Canadian Rockies Hemp	1000.2 g 1.00% 0.10 M Sulfuric acid	185 °C	26 h		4278 g Water	18.9 g	40 °C	1.7%
Comment										
MB3022	AV5	81.8 g Canadian Rockies Hemp	999.9 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4262 g Water	19.0 g	40 °C	2.9%
Comment										
MB3024	AV5	80.7 g Canadian Rockies Hemp	999.6 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4256 g Water	18.4 g	40 °C	2.0%
Comment										
MRa349	AV5	80.8 g Canadian Rockies Hemp	999.4 g 1.00% 0.10 M Sulfuric acid	185 °C	26 h		4254 g Water	19.1 g	40 °C	2.2%
Comment										
MRa354	AV5	80.2 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	185 °C	26 h		4414 g Water	18.4 g	40 °C	1.7%
Comment										
MRa355	AV5	80.2 g Canadian Rockies Hemp	999.5 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4262 g Water	18.2 g	40 °C	1.8%
Comment										
MRa358	AV5	80.8 g Canadian Rockies Hemp	999.9 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		3317 g Water	18.8 g	40 °C	2.1%
Comment										
MB3025	AV5	81.4 g Canadian Rockies Hemp	1000.5 g 1.00% 0.10 M Sulfuric acid	165 °C	21.5 h		4229 g Water	18.4 g	40 °C	1.1%
Comment										
MB3029	AV5	81.6 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	185 °C	26 h		4178 g Water	18.8 g	40 °C	1.7%
Comment										
MB3031	AV5	82.5 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	190 °C	25 h		4178 g Water	19.8 g	40 °C	1.5%
Comment										
MB3032	AV5	81.9 g Canadian Rockies Hemp	1000.5 g 1.00% 0.10 M Sulfuric acid	180 °C	23.5 h		4341 g Water	19.0 g	rt	4.8%
Comment										
MRa367	AV5	81.9 g Canadian Rockies Hemp	1000.0 g 1.00% 0.10 M Sulfuric acid	190 °C	24.5 h		4256 g Water	19.8 g	rt	4.7%
Comment										
MRa368	AV5	81.4 g Canadian Rockies Hemp	999.8 g 1.00% 0.10 M Sulfuric acid	185 °C	27.5 h		4290 g Water	20.3 g	rt	5.5%
Comment										
MB3034	AV5	84.7 g Canadian Rockies Hemp	1000.2 g 1.00% 0.10 M Sulfuric acid	180 °C	24.5 h		4254 g Water	20.0 g	40 °C	2.0%
Comment										
MB3037	AV5	80.7 g Canadian Rockies Hemp	999.8 g 1.00% 0.10 M Sulfuric acid	185 °C	21.5 h		4325 g Water	18.6 g	40 °C	
Comment										

Experiment	Reactor	Raw material	Acid	T	t	pressure	Wash	Output	Drying	KFT
MB3039	AV5	81.2 g Canadian Rockies Hemp	1000.0 g 1.00% 0.10 M Sulfuric acid	180 °C	25 h		4253 g Water	19.1 g	40 °C	
Comment										
MB3040	AV5	80.0 g Canadian Rockies Hemp	1002.2 g 1.00% 0.10 M Sulfuric acid	185 °C	26 h		4301 g Water			
Comment										
MRa3042	AV5	80.0 g Canadian Rockies Hemp	999.9 g 1.00% 0.10 M Sulfuric acid	185 °C	26 h		4026 g Water			
Comment										
MB3044	AV5	82.4 g Canadian Rockies Hemp	1000.0 g 1.00% 0.10 M Sulfuric acid	180 °C	25 h		4316 g Water			
Comment										
MB3045	AV5	82.8 g Canadian Rockies Hemp	1000.1 g 1.00% 0.10 M Sulfuric acid	160 °C	23 h		3964 g Water		rt	
Comment										
MB3047	AV5	81.1 g Canadian Rockies Hemp	1000.1 g 1.00% 0.10 M Sulfuric acid	160 °C	24 h		3836 g Water		rt	
Comment										
MB3048	AV5	80.2 g Canadian Rockies Hemp	1000.2 g 1.00% 0.10 M Sulfuric acid	160 °C	23 h		4275 g Water			
Comment										
MB3049	AV5	Canadian Rockies Hemp	1.00% 0.10 M Sulfuric acid	160 °C	24 h		Water			
Comment										

Comments

- Red text for T and t shows experiments where the temperature tracks have not yet been evaluated (or are not available)

Experiment overview – Step 2

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
TB1131	A	1.2 g	MB2933	1.2 g KOH (90%)	manual	Ar	20-27 °C/min	790 °C	1 h	20 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.16 g
Comment		ground biochar: brown powder, strong electrostatic charge										
TB1132	A	1.3 g	MB2935	1.3 g KOH (90%)	mill (2 min)	Ar	20-29 °C/min	792 °C	1 h	20 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.3 g
Comment		ground biochar: brown powder										
TB1133	A	1.0 g	MRa231	1.0 g KOH (90%)	mill (1 min)	Ar	3 °C/min	785 °C	1 h	20 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.3 g
Comment		ground biochar: brown powder; see temperature trend										
TB1134	A	1.0 g	MRa231	1.0 g KOH (90%)	mill (2 min)	Ar	3 °C/min	771 °C	1 h	20 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.3 g
Comment		ground biochar: brown powder; see temperature trend										
MB2946	A	1.0 g	MB2946X	1.0 g KOH (90%)	mill (1 min)	Ar	5 °C/min	770 °C	1 h	22 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.3 g
Comment		ground biochar: brown powder; see temperature trend										
MB2947A	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	803 °C	1 h	10 g 10% HCl (+ water)	40 °C N2 stream 210 mbar	< 0.01 g
Comment		ground biochar: brown powder; see temperature trend										
MB2947B	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	804 °C	2 h	10 g 10% HCl (+ water)	40 °C N2 stream 210 mbar	< 0.01 g
Comment		ground biochar: brown powder; see temperature trend										
MB2947C	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	806 °C	3 h	Batch discarded		
Comment		ground biochar: brown powder; see temperature trend										
MB2947D	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	810 °C	3 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	38 mg
Comment		ground biochar: brown powder; see temperature trend										
TB1135-1	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	14 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.06 g
Comment		ground biochar: brown powder; see temperature trend										
TB1135-2	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	14 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.07 g
Comment		ground biochar: brown powder; see temperature trend										
MB2952B	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.07 g
Comment		ground biochar: brown powder; see temperature trend										
MB2952C	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	3 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.05 g
Comment		ground biochar: brown powder; see temperature trend										

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
MB2952C2	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2955A	A	0.25 g	MB2946X	0.38 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	7 mg
Comment		Quartz tube shattered during experiment								(+ water)	(atm. Pressure)	
MB2955A2	A	0.25 g	MB2946X	0.38 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
TB1136	A	0.26 g	MB2946X	0.39 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	14 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2962A	A	0.25 g	MB2946X	0.50 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2962B	A	0.25 g	MB2946X	0.50 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2963A	A	0.22 g	MB2960	0.22 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2963B	A	0.29 g	MB2960	0.29 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.12 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
TB1137-1	A	0.26 g	MB2946X	1.06 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	23 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
TB1137-2	A	0.28 g	MB2946X	1.13 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	23 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
TB1137-3	A	0.26 g	MB2946X	1.06 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	23 g 10% HCl	100 °C N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2965A	A	0.22 g	MB2959	1.08 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.02 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend								(+ water)	(atm. Pressure)	
MB2965B	A	0.22 g	MB2959	1.08 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g 10% HCl	100 °C N2 stream	0.01 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend								(+ water)	(atm. Pressure)	
MB2965B2	B	0.22 g	MB2959	1.08 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g 10% HCl	100 °C N2 stream	0.01 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend								(+ water)	(atm. Pressure)	

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
MB2966A	B	0.26 g	MB2959	1.04 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2966B	A	0.26 g	MB2959	1.04 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g 10% HCl	100 °C N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2967A	B	0.26 g	MB2964	1.04 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2967B	A	0.26 g	MB2964	1.04 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g 10% HCl	100 °C N2 stream	0.04 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2970A	B	0.24 g	MB2957	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2970B	A	0.24 g	MB2957	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.02 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2971A	B	0.24 g	MB2958	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2971B	A	0.24 g	MB2958	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2972A	B	0.24 g	MB2961	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2972B	A	0.24 g	MB2961	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2973A	B	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.04 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2973B	A	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2974A	B	0.24 g	MB2959	0.36 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2974B	A	0.24 g	MB2959	0.36 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
MB2975A	B	0.24 g	MB2959	0.36 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2975B	A	0.24 g	MB2959	0.36 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2976A	B	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar (brown powder) compacted to pellet; see temperature trend								(+ water)	(atm. Pressure)	
MB2976B	A	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.11 g
Comment		ground biochar (brown powder) compacted to pellet; see temperature trend								(+ water)	(atm. Pressure)	
MB2979A	B	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2979B	A	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.10 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2980A	B	0.24 g	MB2959/2956	0.36 g KOH (90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
		ground biochar (brown powder) compacted to pellet								(+ water)	(atm. Pressure)	
MB2980B	A	0.24 g	MB2959/2956	0.36 g KOH (90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.10 g
		ground biochar (brown powder) compacted to pellet								(+ water)	(atm. Pressure)	
MB2980C	B	0.52 g	MB2959/2956	0.78 g KOH (90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	21 g 10% HCl	100 °C N2 stream	0.16 g
		ground biochar (brown powder) compacted to pellet								(+ water)	(atm. Pressure)	
MB2980D	A	0.52 g	MB2959/2956	0.78 g KOH (90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	21 g 10% HCl	100 °C N2 stream	0.23 g
		ground biochar (brown powder) compacted to pellet								(+ water)	(atm. Pressure)	
KJo-0165A	A	0.24 g	MB2978	0.36 g KOH (90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.09 g
		ground biochar: brown powder								(+ water)	(atm. Pressure)	
KJo-0165B	B	0.24 g	MB2978	0.36 g KOH (90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	11 g 10% HCl	100 °C N2 stream	0.15 g
		ground biochar: brown powder								(+ water)	(atm. Pressure)	
KJo-0166A	A	0.24 g	MB2978	0.36 g KOH (90%)	mill (10 min)	N2	10 °C/min	800 °C	1 h	11 g 10% HCl	100 °C N2 stream	0.13 g
		ground biochar: brown powder								(+ water)	(atm. Pressure)	
KJo-0166B	B	0.24 g	MB2978	0.36 g KOH (90%)	mill (20 min)	N2	10 °C/min	800 °C	1 h	11 g 10% HCl	100 °C N2 stream	0.11 g
		ground biochar: brown powder								(+ water)	(atm. Pressure)	

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
KJo-0167A	A	0.24 g	MB2978	0.36 g KOH (90%)	mill (10 min)	N2	3 °C/min	800 °C	4 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.10 g
		ground biochar (brown powder) compacted to pellet										
KJo-0167B	B	0.24 g	MB2978	0.36 g KOH (90%)	mill (20 min)	N2	3 °C/min	800 °C	4 h	11 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.06 g
		ground biochar (brown powder) compacted to pellet										
KJo-0168A	A	0.24 g	MB2978	0.36 g KOH (90%)	mill (10 min)	N2	3 °C/min	900 °C	2 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.02 g
		ground biochar (brown powder) compacted to pellet										
KJo-0168B	B	0.24 g	MB2978	0.36 g KOH (90%)	mill (20 min)	N2	3 °C/min	900 °C	1 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.09 g
		ground biochar (brown powder) compacted to pellet										
KJo-0169	B	0.80 g	MB2978	1.20 g KOH (90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	30 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.23 g
		ground biochar (brown powder) compacted to pellet										
KJo-0170	B	1.07 g	MB2978	1.61 g KOH (90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	38 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.62 g
		ground biochar (brown powder) compacted to pellet										
KJo-0171A	B	1.60 g	MB2978	2.40 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	60 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.64 g
		ground biochar (brown powder) compacted to two pellets of equal size										
KJo-0171B	A	0.24 g	MB2978	0.36 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.13 g
		ground biochar (brown powder) compacted to pellet										
KJo-0172A	B	0.50 g	MB2978	0.50 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	11 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.24 g
		ground biochar (brown powder) compacted to pellet										
KJo-0172B	A	0.50 g	MB2978	0.50 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.17 g
		ground biochar (brown powder) compacted to pellet										
KJo-0175	A	1.60 g	MB2978	2.40 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	61 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.71 g
		ground biochar (brown powder) compacted to two pellets of equal size, quartz tube shattered										
KJo-0176	B	0.24 g	MB2978	0.36 g KOH (90%)	mill (5 min)	N2	1.5 °C/min	800 °C	1 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.10 g
		ground biochar (brown powder) compacted to pellet										
KJo-0177	A	1.60 g	MB2978	2.40 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	61 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.91 g
		ground biochar (brown powder) compacted to two pellets of equal size										
KJo-0178	A	1.60 g	MB2955/74/75/76	2.40 g KOH (90%)	mill (1-5 min)	N2	3 °C/min	800 °C	1 h	60 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.55 g
		ground biochar (brown powder) compacted to two pellets of equal size										
MB2981	B	1.44 g	MB2955/74/75/76	2.16 g KOH (90%)	mill (1-5 min)	N2	3 °C/min	800 °C	1 h	60 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g
		ground biochar (brown powder) compacted to six pellets of equal size										

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Volume	Species	Appearance
TB1138	A	1.54 g	various	2.31 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	59 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.59 g 38%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa320	B	1.43 g	various	2.14 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	53 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.55 g 38%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2984	B	1.40 g	various	2.10 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	53 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.48 g 34%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1139	A	1.39 g	KJo-0173	2.08 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	50 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.54 g 39%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1140A	A	1.28 g	KJo-0173	1.92 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	49 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 36%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1140B	B	1.28 g	KJo-0173	1.92 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	50 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.48 g 38%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa323	A	1.06 g	KJo-0173	1.60 g KOH (90%)	mill (3 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g 41%		1	black/grey brittle
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size														
TB1141	A	1.08 g	various	1.62 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	30 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.40 g 37%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa326	A	1.28 g	KJo-0173	1.92 g KOH (90%)	mill (3 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.51 g 39%		1	black/grey brittle
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size														
MRa327	B	1.04 g	KJo-0173	1.56 g KOH (90%)	mill (3 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 44%		1	black/grey brittle
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size														
MB2988	B	1.04 g	KJo-0173	1.56 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	45 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.42 g 40%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa328	B	1.20 g	KJo-0173	1.80 g KOH (90%)	mill (3 min)		N2	3 °C/min	800 °C	1 h	45 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.53 g 44%		1	black/grey brittle
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size														
MRa329	A	1.04 g	KJo-0173 (extra dried)	1.56 g KOH (90%)	mill (1 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.43 g 42%		2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2990A	A	1.28 g	MB2982	1.92 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.50 g 39%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2990B	B	1.04 g	MB2982	1.56 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.42 g 40%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MRa331A	B	1.28 g	MB2985 (extra dried)	1.92 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.64 g 50%		Mostly 1	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa331B	A	1.04 g	MB2985 (extra dried)	1.56 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 44%		Mostly 1	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa333A	A	1.28 g	MB2985 (extra dried)	1.15 g KOH (90%) 0.77 g NaOH (98%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.21 g 17%		2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa333B	B	1.04 g	MB2985 (extra dried)	0.94 g KOH (90%) 0.62 g NaOH (98%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.26 g 25%		2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa334A	A	1.04 g	MB2985 (extra dried)	1.25 g KOH (90%) 0.31 g NaOH (98%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.48 g 46%		Mostly 2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa334B	B	0.88 g	MB2985 (extra dried)	1.06 g KOH (90%) 0.26 g NaOH (98%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	33 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.39 g 45%		Mostly 2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2993A	A	1.28 g	MB2982	1.15 g KOH (90%) 0.77 g NaOH (98%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.22 g 18%		Mostly 2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2993B	B	1.04 g	MB2982	0.94 g KOH (90%) 0.62 g NaOH (98%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.19 g 18%		Mostly 2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2995A	A	1.28 g	MB2983 (extra dried)	1.92 g KOH (90%)	mill (4 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.54 g 42%	3.9 ml 7.3 ml/g	Mostly 1	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2995B	B	1.04 g	MB2983 (extra dried)	1.56 g KOH (90%)	mill (4 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 44%	5.6 ml 12.3 ml/g	1/2 mix	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1144A	A	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (15 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	38 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.61 g 59%	5.0 ml 8.2 ml/g	1 + fibres	black/grey somewhat shiny
		ground biochar (brown powder) NOT compacted														
TB1144B	B	1.28 g	KJo-0174	1.92 g KOH (90%)	ball mill (15 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	54 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.36 g 28%	6.0 ml 16.9 ml/g	2 (?)	black/grey shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2997A	A	1.28 g	MB2985 (extra dried)	1.92 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.55 g 43%	3.6 ml 6.6 ml/g	1	black/grey somewhat shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2997B	B	1.04 g	MB2985 (extra dried)	1.56 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	40 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.45 g 43%	2.9 ml 6.4 ml/g	1	black/grey somewhat shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2999A	A	1.28 g	KJo-0174	1.92 g KOH (90%)	ball mill (25 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.68 g 53%	4.6 ml 6.8 ml/g	Mostly 1/2 mix	shiny black/grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MB2999B	B	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (25 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.40 g 38%	4.6 ml 11.5 ml/g	1 + fibres	shiny black/grey voluminous
		ground biochar (brown powder) NOT compacted														
TB1146	A	1.28 g	MB2982	1.92 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	51 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.52 g 41%	4.0 ml 7.6 ml/g	1	black/grey somewhat shiny
		ground biochar/KOH mix (brown powder, MB2995) dried at 100 °C, then compacted to two pellets of equal size														
TB1147	B	1.08 g	MB2983 (extra dried)	1.62 g KOH (90%)	mill (4 min)		N2	3 °C/min	800 °C	1 h	37 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.36 g 33%	5.0 ml 14.1 ml/g	Mostly 1	black/grey somewhat shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3001A	A	1.28 g	KJo-0174	1.92 g KOH (90%)	ball mill (35 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.49 g 38%	10.6 ml 21.5 ml/g	Mostly 2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3001B	B	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (35 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.48 g 46%	8.2 ml 17.2 ml/g	Mostly 2	shiny black/grey very voluminous
		ground biochar (brown powder) NOT compacted														
MB3003A	A	1.28 g	MB2986 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 80 sec)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.57 g 45%	5.7 ml 9.9 ml/g	Mostly 1	black/grey somewhat shiny
		freshly ground biochar (brown powder) compacted to two pellets of equal size														
MB3003B	B	1.04 g	MB2986 (2% water)	1.56 g KOH (90%)	ball mill (35 Hz, 80 sec)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g 42%	9.0 ml 20.7 ml/g	Mostly 1/2 mix	shiny black/grey voluminous
		freshly ground biochar (brown powder) NOT compacted														
MB3004A	A	1.28 g	MB2986 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 80 sec)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.59 g 46%	17.4 ml 29.4 ml/g	Mostly 2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3004B	B	1.04 g	MB2986 (2% water)	1.56 g KOH (90%)	ball mill (35 Hz, 80 sec)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.41 g 39%	8.4 ml 20.7 ml/g	Mostly 1/2 mix	shiny black/grey voluminous
		ground biochar (brown powder) NOT compacted														
TB1151A	A	1.40 g	KJo-0174	2.10 g KOH (90%)	ball mill (15 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	51 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.53 g 38%	2.5 ml 4.7 ml/g	1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1151B	B	1.08 g	KJo-0174	1.62 g KOH (90%)	ball mill (15 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.45 g 42%	2.5 ml 5.6 ml/g	1	black/grey brittle
		ground biochar (brown powder) NOT compacted														
MB3007A	A	1.20 g	KJo-0174	1.80 g KOH (90%)	ball mill (15 Hz, 30 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.50 g 42%	3.0 ml 6.0 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3007B	B	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (15 Hz, 30 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.43 g 42%	2.9 ml 6.7 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) NOT compacted														
MB3009	A	1.28 g	MB2992	1.92 g KOH (90%)	mill (1 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.41 g 32%	3.0 ml 7.4 ml/g	1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3012A	A	1.28 g	MB2983 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.64 g 50%	10.2 ml 16.0 ml/g	1/2 mix	shiny black/grey voluminous
		ground biochar (brown powder) NOT compacted														

Experiment	Oven	Qty	Lot	Base			grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying		Output	Vol/Dens	Species	Appearance
MB3012B	B	1.04 g	MB2983 (2% water)	1.56 g KOH (90%)			ball mill (15 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.44 g 42%	10.8 ml 24.6 ml/g	Mostly 1/2 mix	shiny black/grey voluminous
		ground biochar (brown powder) NOT compacted																	
MB3013A	A	1.28 g	MB2983 (2% water)	1.92 g KOH (90%)			ball mill (35 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.58 g 45%	4.3 ml 7.5 ml/g	Mostly 1/2 mix	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size																	
MB3013B	B	1.04 g	MB2983 (2% water)	1.56 g KOH (90%)			ball mill (15 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.40 g 38%	12.8 ml 32.2 ml/g	Mostly 2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size																	
MB3015A	A	1.20 g	MB2986 (2% water)	1.80 g KOH (90%)			ball mill (15 Hz, 30+1 min)	no	N2	3 °C/min	800 °C	1 h	45 g 10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.39 g 33%	8.0 ml 20.5 ml/g	Mostly 2	shiny black/grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size																	
MB3015B	B	1.00 g	MB2986 (2% water)	1.50 g KOH (90%)			ball mill (15 Hz, 30+1 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.36 g 36%	8.0 ml 22.5 ml/g	Mostly 2	shiny black/grey voluminous
		ground biochar (brown powder) NOT compacted																	
MRa340A	A	1.28 g	MB2996	1.92 g KOH (90%)			ball mill (15 Hz, 120 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.49 g 38%	4.0 ml 8.2 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size																	
MRa340B	B	1.04 g	MB2996	1.56 g KOH (90%)			ball mill (15 Hz, 120 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.44 g 42%	3.5 ml 7.9 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) NOT compacted																	
TB1155A	A	1.28 g	MB2986 (2% water)	1.54 g KOH (90%)	0.38 g NaOH (98%)		ball mill (15 Hz, 10 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.69 g 54%	4.0 ml 5.8 ml/g		black/grey somewhat shiny
		NaOH not fully ground!, ground biochar (brown powder) compacted to two pellets of equal size																	
TB1155B	B	1.04 g	MB2986 (2% water)	1.25 g KOH (90%)	0.31 g NaOH (98%)		ball mill (15 Hz, 10 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.45 g 43%	4.0 ml 8.9 ml/g		black/grey somewhat shiny
		NaOH not fully ground!, ground biochar (brown powder) NOT compacted																	
MB3017A	A	1.28 g	MB3002 (2% water)	1.54 g KOH (90%)	0.38 g NaOH (98%)		ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.53 g 41%	19.1 ml 36.3 ml/g	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size																	
MB3017B	B	1.04 g	MB3002 (2% water)	1.25 g KOH (90%)	0.31 g NaOH (98%)		ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.33 g 32%	12.2 ml 36.7 ml/g	2	shiny black/grey voluminous
		ground biochar (brown powder) NOT compacted																	
MB3019A	A	1.28 g	MB3002 (2% water)	1.92 g KOH (90%)			ball mill (15 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.47 g 37%	15.9 ml 33.9 ml/g	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size																	
MB3019B	B	1.04 g	MB3002 (2% water)	1.56 g KOH (90%)			ball mill (15 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.39 g 38%	9.3 ml 23.7 ml/g	Mostly 2	shiny black/grey voluminous
		ground biochar (brown powder) NOT compacted																	
MB3021A	A	1.28 g	MB3002 (2% water)	1.92 g KOH (90%)			mill (2 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.52 g 41%	7.3 ml 14.0 ml/g	Mostly 1	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size																	
MB3021B	B	1.04 g	MB3002 (2% water)	1.56 g KOH (90%)			mill (2 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.43 g 42%	5.3 ml 12.2 ml/g	Mostly 1	shiny black/grey
		ground biochar (brown powder) NOT compacted																	

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MB3023A	A	1.08 g	MB2989	1.30 g KOH (90%) 0.32 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.28 g 26%	16.5 ml 59.1 ml/g	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3023B	B	0.80 g	MB2989	0.96 g KOH (90%) 0.24 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.25 g 31%	13.5 ml 54.7 ml/g	Mostly 2	shiny black/grey very voluminous
		ground biochar (brown powder) NOT compacted														
MRa352A	A	1.28 g	MB3002 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.62 g 48%	2.9 ml 4.7 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa352B	B	1.04 g	MB3002 (2% water)	1.56 g KOH (90%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.53 g 51%	2.8 ml 5.3 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) NOT compacted														
MB353A	A	1.28 g	MB2991	1.92 g KOH (90%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.53 g 41%	2.0 ml 3.8 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa353B	B	1.04 g	MB2991	1.39 g KOH (90%) 0.17 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g 43%	3.5 ml 7.9 ml/g	Mostly 1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1158A	A	1.04 g	MB2991	1.25 g KOH (90%) 0.31 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	50 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.40 g 39%	17.0 ml 42.3 ml/g	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1158B	B	0.96 g	MB2991	1.37 g KOH (90%) 0.08 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	45 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.39 g 40%	3.0 ml 7.8 ml/g	Mostly 1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1160A	A	1.04 g	MB2991	1.25 g KOH (90%) 0.31 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 44%	17.0 ml 37.0 ml/g	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1160B	B	1.04 g	MB2991	1.56 g KOH (90%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	41 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.43 g 41%	2.0 ml 4.6 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1162A	A	1.04 g	MB2989	1.56 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	45 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g 43%	4.0 ml 9.0 ml/g	Mostly 1	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1162B	B	1.04 g	MB2989	1.25 g KOH (90%) 0.31 g NaOH (98%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	45 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.32 g 31%	8.0 ml 25.2 ml/g	Mostly 2	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3026	B	1.04 g	MB3005 (2% water)	1.56 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.40 g 39%	8.3 ml 20.6 ml/g	1/2 mix	shiny black/grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3027	B	1.04 g	MB3005 (2% water)	1.25 g KOH (90%) 0.31 g NaOH (98%)	mill (3 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.28 g 27%	18.0 ml 64.7 ml/g	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3028	B	1.04 g	MB2989	1.56 g KOH (90%)	mill (10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.42 g 40%	5.1 ml 12.1 ml/g	Mostly 1	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size														

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MB3030	A	1.04 g	MB2989	1.25 g KOH (90%) 0.31 g NaOH (98%)	mill (10 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.29 g 28%	16.8 ml 57.7 ml/g	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1165	C	12.4 g	MB2992	18.5 g KOH (90%)	5 millings (3 min each)	yes	N2	3 °C/min	800 °C	1 h	193 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	4.55 g 37%	35.0 ml 7.7 ml/g	1	dull black/grey
		Rotating oven, powder not compacted														
TB1166	C	7.8 g	MB3006 (2% water)	9.41 g KOH (90%)	4 millings (3 min each)	yes	N2	3 °C/min	800 °C	1 h	210 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	3.25 g 41%	65.0 ml 20.0 ml/g	1/2 mix	shiny black/grey
		Rotating oven, powder not compacted														
MB3036	B	1.04 g	MB2992 + H2O	1.56 g KOH (90%)	mill (1+3 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.34 g 32%	3.0 ml 8.9 ml/g	1	black/grey barely shiny
		water added to mill => 7-8% (calc), ground biochar (brown powder) compacted to two pellets of equal size														
TB1167	C	14.8 g	MB2994	22.3 g KOH (90%)	blender, 90 sec	yes	N2	3 °C/min	800 °C	1 h	200 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	6.40 g 43%	50.0 ml 7.8 ml/g	Mostly 1	shiny black/grey
		Rotating oven, powder not compacted														
MB3038	B	1.04 g	MB2994	1.56 g KOH (90%)	blender, 90 sec	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.53 g 51%	2.4 ml 4.5 ml/g	Mostly 1	black/grey barely shiny
		ground biochar (brown powder from TB1167) compacted to two pellets of equal size														
MRa373	A	1.04 g	MB2989 + H2O	1.56 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.43 g 41%	3.5 ml 8.2 ml/g	Mostly 1	shiny black/grey
		water added to biochar => 8% (KFT), ground biochar (brown powder) compacted to two pellets of equal size														
MRa375	A	1.04 g	MB2996 + H2O	1.56 g KOH (90%)	mill (1+3 min)	yes	N2	3 °C/min	800 °C	1 h	40 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.51 g 49%	3.2 ml 6.3 ml/g	Mostly 1	black/grey barely shiny
		water added to mill => 9% (calc), ground biochar (brown powder) compacted to two pellets of equal size														
MRa376	C	16.7 g	MB2998 + H2O	25.0 g KOH (90%)	blender, 10+80 sec	?	N2	3 °C/min	800 °C	1 h	189 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	7.6 g 46%	60.0 ml 7.9 ml/g	Mostly 1	shiny black/grey
		water added to blender => 7-8% (calc), rotating oven, powder not compacted														
MB3041	B	1.3 g	MB2996	1.3 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 35%	3.0 ml 6.5 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3043	C	16.5 g	MB3000	24.72 g KOH (90%)	blender, 90 sec	yes	N2	3 °C/min	800 °C	1 h	285 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	7.11 g 43%	70.0 ml 9.8 ml/g	Mostly 1	black/grey barely shiny
		Rotating oven, powder not compacted														
MRa380	A	1.0 g	MB2996 + H2O	1.0 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.48 g 46%	5.0 ml 10.4 ml/g	1/2 mix	shiny black/grey
		water added to biochar => 11.5% (KFT), ground biochar (brown powder) compacted to two pellets of equal size														
MB3046	C	15.6 g	MB3032	24.20 g KOH (90%)	blender, 90 sec	yes	N2	3 °C/min	800 °C	1 h	311 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	7.05 g 45%	95.0 ml 13.5 ml/g	*	shiny black/grey
		KOH not milled, rotating oven, powder not compacted														
MRa385	C	17.8 g	MRa367	26.6 g KOH (90%)	blender, 6x15 sec	no	N2	3 °C/min	800 °C	1 h	183 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	8.60 g 48%	125.0 ml 14.5 ml/g	Mostly 1/2 mix	shiny black/grey
		Rotating oven, powder not compacted														
MRa386	A	2.1 g	MRa368	2.1 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	60 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.65 g 32%	3.7 ml 5.7 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														

Comments

- Yellow/red text for the lot of biochar show cases where the quality of the milled material was somewhat/significantly impaired by contact with air and/or moisture (leading to heating or colour change of the material, stickiness, clumping or similar changed in appearance)
- The information on homogeneity of the milled material is based on optical and acoustic evaluation